

UAV traffic management to create a ground for operational safety and efficiency

The UTM policy (Unmanned Aircraft System Traffic Management), once implemented, will be paving the way for operational scalability and safety by creating a carefully designed SOP (Standard Operating Protocol) to be followed by the start-ups.

The Digital Sky Platform is expected to be acting as traffic regulator, that can be effective in managing unmanned aircraft traffic based on 'no permission, no takeoff' (NPNT) policy, while helping the operators acquire licenses (UIN - Registration/UAOP - Driving License) with minimum effort.

The Drone Policy 2.0 also accounts for thorough monitoring of drones—for instance, monitoring how drones collect, use, store, and share data.

Standardisation to be driven by the government policy, that can lead to commercialisation and hence can attract the investors. Rahul Verma, an operational RPA instructor, thinks that the operational challenges can be overcome through the integration of ATC (Air Traffic Controller) with the UTC, thus integrating UAV and pilots with the existing airspace being controlled and managed by ATC.

He suggests that it requires a blend of operational and technological improvements. It involves new set of requirements of ATC to manage UAV that is scalable and should be used to develop an information sharing methodologies (sharing between operators and UTC). Accordingly, a workflow for RPA/UAV mission is required to standardise the operations with minimum acceptable flight risk. In short, UTC can be treated as a sub-system within a larger system of ATC.

Communication technology

The business viability of Drone operators in significant number of applications can only be established when the drone can be flown Beyond the Visual Line of Sight (BVLOS), or when it flies autonomously, in order to justify the operating cost involved and the ticket size of the projects.

In such cases of BVLOS flights,

when the data is transmitted to the ground stations beyond the remote-control station of the pilot it becomes the key success factor. This necessitates the drone manufacturers and the telecom infrastructure companies to invest in the hardware to meet following requirement:

- The BVLOS drones should be equipped with the mobile communication modules
- A strong network coverage through the 5G technology will be paramount in order to support the communication requirement
- The telecom industry should work in tandem with the drone industry to ensure availability of the coverage and bandwidth up to the lower part of the airspace

Training centres are emerging....

Services being at the centre stage of delivery of drone solutions, the pilot training too is expected to gain prominence, which is mandatory as per the DGCA norm, in order to get the licenses to fly drones for commercial purpose.

This, in turn, is going to encourage many drone technology providers and fleet management companies to set up training institutes to cater to the market demand in coming 2-3 years. The trained pilots will be playing a crucial role in providing operational sustainability to the ecosystem. Many drone manufacturers in India, including Aarav Unmanned Systems, Aero360, Johnnet Technologies, and Skyport innovations, are offering training solutions to the enterprises and individuals.

Drone operators and system integrators to play a key role in providing the scale through innovative business models

Data analytics and innovative business models offering flexible pricing are going to be the key differentiating factors. There are generalist drone operators for regular use cases such as cinematography, photo shooting, etc,

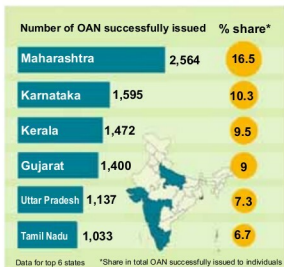


Fig. 6: Number of OAN successfully issued (Source: Business Today.in)

and there are specialised geo-spatial solution providers that can include geospatial video survey and inspection to be used for 3D Modeling and volumetric analysis, etc.

In the wake of rapid adoption of drone technology, Indian entrepreneurs are showing keen interest in grabbing a pie in the continuously evolving value chain. A host of business models are taking shape as the demand for UAVs is accelerating. While drone manufacturers are key proponents of the technology, a number of variants of service models are expected to emerge including rental services, drone-as-a-service, system integrators offering specialised services, surveillance-as-a-service, etc.

Preferred business models for the investors

Seventy per cent of the start-ups can get funding subject to right business strategy and resources.

Experts suggest that the start-ups initially should focus on one idea and its successful execution. Ability to specialise will attract investors, who look for the market, expertise, differentiated disruptive capability, team capability, etc. There can be specific funding for various use cases, which necessitates to build differentiated lab facility and, most important, to build inclusive ecosystem.

Drone services are emerging out to be the most preferred business model among the start-ups on the backdrop of high scope of business scalability. The